METRIC

MIL-N-82881
27 October 1993
SUPERSEDING
MIS-46256
18 December 1991

MILITARY SPECIFICATION

NITROSTARCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for a wet nitrostarch for the base charges of small explosive devices. This nitrostarch is of one type and one class and is prepared by nitrating a starch from which excess fats and pectin acid have been removed.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1376

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation (October 1, 1989 revision)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2905	Standard Practice for Statements on Number of Specimens for Textiles

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Manufacturing requirement. The nitrostarch shall be manufactured from cornstarch which, prior to nitrating, is chemically treated to remove excess fats and pectic acid.

3.2 Chemical requirements. The nitrostarch shall conform to the chemical requirements of table I when tested as specified.

TABLE I. Chemical requirements.

Requirement	Value	Test Paragraph
Water content, % by weight, minimum (M101)	20	4.6.1
Nitrogen content, % by weight (M102)	12.75 - 13.00	4.6.2
Acetone insoluble material, % by weight, maximum (M103)	0.40	4.6.3
Sulfates, % by weight, maximum (M104)	0.05	4.6.4
Ash, % by weight, maximum (M105)	0.40	4.6.5
Heat stability at 134.5°C, time in minutes, minimum (M106)	50	4.6.6

3.3 Physical requirements. The nitrostarch shall conform to the physical requirements of table II when tested as specified.

TABLE II. Physical requirements.

Requirement	Value	Test Paragraph
Form (M107)	Non-caking powder	4.6.7
Color (M108)	Light tan	4.6.7
Loss of weight at 100°C, %, maximum (M109)	2.0	4.6.8
Ignition temperature, minimum (M110)	150°C	4.6.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3)

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. A lot shall consist of one or more batches of nitrostarch produced by one manufacturer, in accordance with the same specification under one continuous set of operating conditions. Each lot shall consist of that quantity of nitrostarch that has been subjected to the same unit chemical and/or physical process intended to make the final product homogeneous. The lot shall comply with the provisions for submission of product samples as specified in MIL-STD-105. Lot number assignment shall be in accordance with MIL-STD-1168. When specified in the contract or purchase order (see 6.2), the provisions of MIL-A-48078 shall apply.

4.3.2. Sample size. Quality conformance inspection sample size shall be as specified in the contract or purchase order (see 6.2).

4.3.3 Inspection. Quality conformance inspection lots or samples shall be subjected to the tests and examinations of 4.6.

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements not annotated with a classification code are classified as minor.

4.5 Testing.

CAUTION

This specification covers sampling and testing of toxic and hazardous material. Accordingly, all applicable safety rules, regulations and procedures must be followed in handling and processing the nitrostarch. Disposal of all chemicals and hazardous materials must be in accordance with all local, state, and federal regulations.

4.5.1 Inspection equipment. Inspection equipment and its calibration, design and approval shall be as specified in the contract or purchase order (see 6.2).

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4.6 Test procedure. All tests shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Unless otherwise specified, all chemicals shall be reagent grade or ACS grade. It is recommended that all test procedures be documented by Standard Operating Procedures (SOPs) for organizational use. (See 6.3 for use of equivalent test methods.)

4.6.1 Water content test. Accurately weigh 5.000 ± 0.500 grams (g) of nitrostarch into an appropriate tared container, such as a 2.5-inch diameter aluminum cup or 4-inch watch glass. Then dry this sample over concentrated sulfuric acid in a desiccator to constant weight at room temperature. Approximately 48 hours is required to dry a sample of this size. The percent loss in weight of the sample shall be determined and recorded as percent water content. The percent by weight of water content shall conform to the value specified in table I.

4.6.2 Nitrogen content test. The nitrogen content shall be determined using a nitrometer on a 1.0 g specimen of the dried nitrostarch used in 4.6.1. Testing of the specimen shall be in accordance with Method 209.3.2 of MIL-STD-286. The percent by weight of nitrogen content shall conform to the value specified in table I.

4.6.3 Acetone insoluble material test. A 1.0 g specimen of the dried nitrostarch used in 4.6.1 shall be weighed to the nearest 0.1 milligram (mg) and transferred to a 250-milliliter (ml) Erlenmeyer flask. The material shall be moistened with 10 ml of ethyl alcohol. A 150 ml portion of filtered, reagent grade acetone shall be added and this mixture shaken until solution is complete. The solution shall be filtered through a well prepared and tared Gooch crucible. Any residue on the flask shall be transferred to the Gooch crucible by means of several small rinses of acetone. The excess of acetone in the crucible shall be removed by sucking air through for several seconds. The crucible shall be dried for 45 minutes in an oven maintained at 100 to 105 degrees Celsius (°C). Any increase in weight of the crucible shall be calculated as percentage of acetone insoluble materials. The percent by weight of acetone insoluble material shall conform to the value specified in table I.

4.6.4 Sulfates test. Weigh 20.00 ± 0.10 g of the wet material and transfer it to a 250-ml beaker. Add 20 ml of reagent grade hydrochloric acid (sulfate free) to the beaker. Cover with a watch glass and digest on the steam bath for 1 hour. Add 75 ml of water to the beaker and continue heating on the steam bath for 1/2 hour. Filter the contents of the beaker through a #1 Whatman filter, and wash the residue on the filter 10 times with small portions of hot water. Collect the filtrate and washings in a 400-ml beaker. Dilute this solution to about 200 ml and bring to a boil on a hot plate. Add 10 ml of 10% $BaCl_2$ solution slowly by means of a pipette to the boiling sample. Digest solution and precipitate on the steam bath for 3 to 6 hours. Then filter while hot on a #42 Whatman filter. Transfer all of the precipitate to the filter and wash well (6 to 8 times) with hot water. Transfer the filter paper and precipitate to a freshly ignited and tared crucible. Dry the crucible and contents in an oven (70° to 100°C for 1 hour) and then ignite at 800°C in a muffle. Cool the crucible and contents in a desiccator and weigh.

$$\% \text{ Sulfates as } H_2SO_4 = P \times \frac{42.02}{W}$$

where

P = Weight of precipitate, g
W = Dry weight of sample, g (calculated; see 4.6.1)

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The percent by weight of sulfates shall conform to the value specified in table I.

4.6.5 Ash test. The percent of ash shall be determined on a 1.0 g specimen taken from a sample of nitrostarch dried in accordance with 4.6.1. Place the dried specimen into a previously ignited and subsequently tared crucible, and weigh to the nearest 0.1 mg. Add, with gentle stirring, sufficient acetone (containing 5 percent by volume of castor oil (see 4.6.5.1)) to gelatinize the nitrostarch (see 4.6.5.2). The crucible shall be gently ignited and then the nitrostarch shall be allowed to burn, without applying any further heat, until a charred residue remains. The ignition temperature shall be continued until no further change is noted in the contents. The crucible shall be cooled in a desiccator and reweighed. The gain in weight of the crucible shall be calculated as percent ash. The percent by weight of ash shall conform to the value specified in table I.

4.6.5.1 Blank determination. A blank shall be run concurrently with the specimen on the same volume of reagents employed. The blank shall be ignited in the same manner as the specimen. Any residue of ash found in the blank shall be subtracted from the ash found in the nitrostarch specimen.

4.6.5.2 Digesting of nitrostarch. If preferred, the nitrostarch may be digested in the crucible with nitric acid and then ignited directly. Avoid spattering of liquids while heating. Run a blank on reagents as specified in 4.6.5.1.

4.6.6 Heat stability at 134.5°C test. Heat stability of the nitrostarch shall be determined using 2.5 g samples. Each sample shall be tested in accordance with Method 404.1.2 of MIL-STD-286 at a temperature of 134.5°C. Record the time at which the methyl violet test paper has completely changed to salmon pink. The recorded time shall conform to the value specified in table I.

4.6.7 Form and color tests. The nitrostarch shall be examined visually to determine the form and color. The form and color shall conform to the descriptions in table II.

4.6.8 Loss of weight at 100°C test. From a sample of dried nitrostarch (see 4.6.1), place an approximately 0.6 g specimen in a tared test tube 80 mm long X 10 mm inside diameter. Weigh the test tube with contents to the nearest 0.1 mg and, subsequently, heat the tube in a constant temperature bath at 100.0 ± 0.2°C for 48 hours. Reweigh the test tube and determine the loss in weight of the specimen. The percent loss in weight shall conform to the value specified in table II.

4.6.9 Ignition temperature test. From a sample of dried nitrostarch (see 4.6.1), weigh 0.100 ± 0.005 g in each of two pyrex glass test tubes. Immerse the test tubes in a Wood's metal bath to a depth of 0.5 inch. Between these two test tubes position a third test tube containing a thermometer. From 100°C, the temperature of the Wood's metal bath is raised at the rate of 4° to 5°C per minute until deflagration or explosion occurs. The temperature reading of the thermometer when explosion occurs is the ignition temperature. The ignition temperature shall conform to the value specified in table II.

4.7 Inspection of packaging. The packaging, packing and marking shall be examined visually for conformance to section 5.

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5. PACKAGING

5.1 Packaging and packing. The nitrostarch shall be shipped in the wet condition (containing not less than 20 percent by weight of water, see table I). The nitrostarch shall be packaged and packed in accordance with 49 CFR 173.184 (October 1, 1989 revision).

5.2 Marking. Unless otherwise specified (see 6.2), all markings shall be legible and in accordance with MIL-STD-129 and 49 CFR 173.153 (October 1, 1989 revision).

6. *NOTES*

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nitrostarch is intended for use in small explosive devices of fuzes.

6.2 Acquisition requirements. *Acquisition documents must specify the following:*

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Amount of nitrostarch to be procured.

d. The agency that is to perform the inspection (see 4.1).

e. Whether the provisions of MIL-A-48078 apply (see 4.3.1).

f. Quality conformance inspection lot and sample size (see 4.3.1 and 4.3.2).

g. Inspection equipment, calibration, design and approval (see 4.5.1 and MIL-STD-45662).

h. Marking, if other than as specified (see 5.2).

i. Safety precautions (see 4.5 and 6.4).

6.3 Equivalent test methods. The test methods specified herein are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (in accuracy and precision) to the method given herein. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer. This description should include the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

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6.4 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3.

NOTE: When this document is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.5 Subject term (key word) listing.

Base charges
Explosive device
Fuzes

Custodians:
Army-AR
Navy-OS

Preparing activity:
Navy-OS
(Project 1376 -0463)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-N-82881	2. DOCUMENT DATE (YYMMDD) 931027

3. DOCUMENT TITLE
NITROSTARCH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	b. TELEPHONE (Include Area Code) (1) Commercial: 301-743-4358/4510 (2) AUTOVON: 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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